

Supplementary material for Two-stage MILP scheduling and structure-aware learning-enhanced branch-and-cut for dual-source mixed-flow rotating-chamber cluster tools

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1 Computational evidence design

The computational study separates formulation validation, nominal method comparison, mechanism-specific controls, manufacturing-factor analysis, parameter sensitivity, and distribution-shift evaluation. Table 1 gives the complete prespecified design. A run is identified by the instance, method, solver seed, training seed where applicable, and experimental profile. All observations are retained, including time-limited runs.

Table 1: Prespecified computational suites.

Suite	Design	Runs
Validation	Four small physical cases, one exact solve each	4
Nominal	20 instances; two solver baselines with five solver seeds and five learned configurations with five solver and five training seeds	2700
Two-stage stability	20 instances, three refinement profiles, five solver seeds, and five training seeds	1500
Factorial DOE	$4 \times 5 \times 3$ wafer-count, recipe-composition, and processing-scale cells with five solver seeds	300
SPBS	48 instances, automatic versus no warm start, ten matched solver seeds	960
Sensitivity	Eight one-factor settings with ten solver seeds	80
Distribution shift	Nine out-of-distribution instances; two solver baselines and five learned configurations with matched solver and training realizations	729
Total	Excluding the four validation cases	6269

All 6269 formal benchmark runs and four validation runs completed without execution failure. Every saved incumbent used in the reported statistics was checked independently against the corresponding mathematical instance. The distribution-shift suite contains 729 retained time-limited runs; every run returns an independently valid incumbent.

2 Indexed MILP specification

This section records the principal indexed constraint families used to instantiate the finite-horizon model; it is a compact mathematical map of the implemented formulation rather than a claim that every generated row is printed here. Let $\mathcal{W}, \mathcal{P}, \mathcal{C}, \mathcal{K}, \mathcal{O}$, and $\mathcal{R}$ denote product wafers, same-mode product-pair groups, rotary chambers, candidate recipe slots, operations, and unit-capacity resources. A slot $k \in \mathcal{K}(p)$ is a chamber–batch–side tuple in 4×1 mode or a chamber–chain-position tuple in 2×2 mode. $\mathcal{A}$ is the set of technological predecessor arcs, and $\mathcal{O}_r$ contains operations that may use resource r . For each chamber c , $\mathcal{V}_c = (v_{c1}, \dots, v_{cnc})$ is the ordered set of candidate processing cycles and $\mathcal{E}_c = \{1, \dots, E_c\}$ is the ordered set of potential cleaning epochs. The parameter H is the maximum number of active processing cycles permitted between two cleans. Furthermore, $\mathcal{J}_c^{\text{DW}}$ contains dummy-wafer use demands and $\mathcal{T}_c$ contains the reusable physical vacuum-zone dummy wafers.

The binary variables are assignment x_{pk} , position activation u_k , operation activation b_o , cycle-to-epoch assignment a_{vce} , epoch use z_{ce} , dummy identity $g_{j\tau}$, and resource order $y_{oo'}^r$. Here b_v is the activation variable b_o for the processing operation associated with cycle $v \in \mathcal{V}_c$, and $z_{ce} = 1$ means that epoch e of chamber c contains at least one active cycle. Likewise, b_j below abbreviates the activation of dummy-use demand j . Continuous variables s_o, e_o are event start and completion times.

Pairs belonging to one recipe mode have identical processing data and their identifiers carry no physical meaning. They are therefore ordered by identifier and mapped to the compatible slots in a fixed interleaved chamber order. Define the canonical incidence

$$\bar{x}_{pk} = \mathbf{1}\{k = k^{\text{can}}(p)\}, \quad x_{pk} = \bar{x}_{pk} \quad (p \in \mathcal{P}, k \in \mathcal{K}(p)). \quad (1)$$

Equation 1 is a same-mode symmetry reduction: it chooses one representative from each orbit of pair-label permutations and does not remove a physically distinct schedule under the modeled identical-pair data. The odd product/dummy tail, if present, occupies the last compatible 2×2 position. The resulting fixed-duration, activation, and assignment relations are

$$\sum_{k \in \mathcal{K}(p)} x_{pk} = 1 \quad (p \in \mathcal{P}), \quad (2)$$

$$x_{pk} \leq u_k, \quad u_{k+1} \leq u_k, \quad b_o = \chi_o(x, u, a) \quad (o \in \mathcal{O}), \quad (3)$$

$$b_{o'} \leq b_o \quad ((o, o') \in \mathcal{A}), \quad (4)$$

$$0 \leq s_o \leq Mb_o, \quad e_o = s_o + d_o b_o \quad (o \in \mathcal{O}), \quad (5)$$

where the binary affine incidence map χ_o activates exactly the events belonging to the selected 4×1 side, 2×2 chain position, or cleaning cycle. The predecessor inequality makes operation activation consistent along every generated technological chain: an active successor cannot have an inactive required predecessor. The route logic is generated from the following active event chains:

Structure	Ordered active events
Product route	load port $\rightarrow$ ATR $\rightarrow$ aligner $\rightarrow$ upper load lock $\rightarrow$ VTR load $\rightarrow$ chamber process $\rightarrow$ VTR unload $\rightarrow$ lower load lock $\rightarrow$ ATR $\rightarrow$ load port
4×1	front load $\rightarrow$ rotate/back load $\rightarrow$ two-side process $\rightarrow$ back unload $\rightarrow$ rotate/front unload
2×2	dummy/product head $\rightarrow$ adjacent product bridges $\rightarrow$ product/dummy tail; consecutive positions share the prescribed product pair and may split only at a cleaning boundary
Cleaning	front dummy load $\rightarrow$ back dummy load $\rightarrow$ clean process $\rightarrow$ back unload $\rightarrow$ front unload

For every active technological arc and every unordered pair competing for a resource, timing feasibility is imposed by

$$s_{o'} \geq e_o + \delta_{oo'} - M(2 - b_o - b_{o'}) \quad (o, o') \in \mathcal{A}, \quad (6)$$

$$s_{o'} \geq e_o + \rho_{oo'}^r - M(1 - y_{oo'}^r) - M(2 - b_o - b_{o'}), \quad (7)$$

$$s_o \geq e_{o'} + \rho_{o'o}^r - M y_{o'o}^r - M(2 - b_o - b_{o'}) \quad \{o, o'\} \subseteq \mathcal{O}_r. \quad (8)$$

The setup $\rho_{oo'}^r$ includes the empty reposition time from the endpoint of o to the origin of o' . These disjunctions cover the atmospheric and vacuum robots, aligner, individual load-lock slots, rotary chambers, and physical dummy-wafer identities.

Dummy identity and cleaning are linked by

$$\sum_{\tau \in \mathcal{T}_c} g_{j\tau} = b_j \quad (j \in \mathcal{J}_c^{\text{DW}}), \quad (9)$$

$$\sum_{e \in \mathcal{E}_c} a_{vce} = b_v, \quad a_{vce} \leq z_{ce}, \quad z_{ce} \leq \sum_{v \in \mathcal{V}_c} a_{vce}, \quad (10)$$

$$\sum_{v \in \mathcal{V}_c} a_{vce} \leq H z_{ce}, \quad z_{c,e+1} \leq z_{ce}, \quad (11)$$

$$b_{v_{c,q+1}} \leq b_{v_{cq}}, \quad (12)$$

$$\sum_{e \in \mathcal{E}_c} e a_{v_{cq},c,e} \leq \sum_{e \in \mathcal{E}_c} e a_{v_{c,q+1},c,e} + E_c(1 - b_{v_{c,q+1}}) \quad (q = 1, \dots, n_c - 1). \quad (13)$$

If $g_{j\tau} = g_{j'\tau} = 1$, the same resource disjunction is activated for dummy uses j and j' , preventing overlapping reuse. Epoch transitions activate the intervening dummy-only cleaning route, with its activation equal to $z_{c,e+1}$. The two last inequalities impose prefix activation and nondecreasing epoch indices on consecutive active cycles; the term $E_c(1 - b_{v_{c,q+1}})$ prevents an inactive suffix from falsely constraining its active predecessor. The adjacency equations in χ_o permit a 2×2 split only at an activated epoch transition.

Residence and completion constraints are

$$0 \leq s_{\text{out}(w,r)} - e_{\text{in}(w,r)} \leq \bar{R}_{wr}, \quad (14)$$

$$C_{\max} \geq e_w^{\text{return}} \quad (w \in \mathcal{W}). \quad (15)$$

Stage one minimizes $C_{\max}$. Stage two fixes the retained routing and resource-order integers from its stage-one incumbent (double-gripper choices remain free), adds $C_{\max} \leq \hat{C} + 10^{-6}$, and minimizes

$$\Phi = L_{\text{sum}} + 10L_{\max} + 25D_{\text{cadence}} + 0E_{\text{cap}}. \quad (16)$$

Let $w_i \geq 0$ denote a model-native wait or capacity-adjusted idle term, $i \in \mathcal{L}$, let $\bar{w}_i$ be its optional soft target, and let h_{mcel} be the ℓ th active processing start of mode m , chamber c , and epoch e . The max, absolute-value, and optional soft-cap terms are linearized by

$$L_{\text{sum}} = \sum_{i \in \mathcal{L}} w_i, \quad L_{\max} \geq w_i \quad (i \in \mathcal{L}), \quad (17)$$

$$\Delta_{mcel} = (h_{mce,\ell+1} - h_{mce,\ell}) - (h_{mce,\ell} - h_{mce,\ell-1}), \quad (18)$$

$$d_{mcel} \geq \Delta_{mcel}, \quad d_{mcel} \geq -\Delta_{mcel}, \quad D_{\text{cadence}} = \sum_{m,c,e,\ell \geq 2} d_{mcel}, \quad (19)$$

$$\xi_i \geq w_i - \bar{w}_i, \quad \xi_i \geq 0, \quad E_{\text{cap}} = \sum_{i \in \mathcal{L}} \xi_i. \quad (20)$$

The symbol ξ_i is reserved for cap excess and is distinct from the epoch indicator z_{ce} . A positive coefficient makes each epigraph variable tight at optimum. The formal profile disables the soft target, so its effective cap weight is zero and only the first three terms enter the reported objective. Hence stage two preserves the stage-one makespan within numerical tolerance; if stage one is time-limited, this statement applies to its retained incumbent rather than to an unproved global optimum.

For the deterministic offline problem, a scheduling deadlock is a circular wait among active material-transfer or processing operations in which every operation awaits a resource or predecessor held ahead of it. Such a wait would be a directed cycle in the combined technological and selected resource-order graph. Summing the positive separation inequalities around that cycle gives $s_o \geq s_o + \epsilon$ for some $\epsilon > 0$, a contradiction; terminal constraints also require every product and reusable dummy route to finish. Thus every feasible solution is a deadlock-free complete offline trajectory for the modeled resources, durations, and availability assumptions. This is not a runtime deadlock-recovery guarantee under equipment faults, stochastic delays, operator intervention, or any resource state omitted from the model.

3 Independent validation and numerical tolerance

The four small validation cases exercise pure 4×1 , pure 2×2 , and two mixed configurations. Their saved schedules are reloaded, the reported primary decisions are fixed, and the active physical model is checked independently of the learning policy. The same solution-level procedure is then applied to every saved benchmark incumbent. It covers route and stage order, chamber and slot exclusivity, robot and load-lock use, recipe grouping, cleaning, residence limits, and vacuum-zone dummy-wafer capacity. Continuous residuals use the solver tolerance 10^{-6} plus a 10^{-12} serialization allowance, i.e., an effective threshold of 1.000001×10^{-6} . Binary and integer requirements are not relaxed. This allowance only accounts for the decimal text round trip of an otherwise accepted solver solution.

The validation claim is deliberately bounded: it establishes consistency between the generated instances, serialized solutions, and implemented constraints for the tested cases. It is not field validation of unmodeled equipment behavior, a proof of correctness for every possible parameterization, or evidence of online recovery from faults and stochastic disturbances.

4 Event-time reconstruction of schedule stability

Schedule-stability metrics are reconstructed from physical event times rather than auxiliary objective variables. This makes the definition identical under all three refinement profiles. For product wafer w , define six nonnegative route waits:

$$q_{w,1} = [s_w^{\text{PM}} - e_w^{\text{VTRload}}]_+, \quad q_{w,2} = [s_w^{\text{VTRunload}} - e_w^{\text{PM}}]_+, \quad (21)$$

$$q_{w,3} = [s_w^{\text{AL}} - e_w^{\text{ATR,LP} \rightarrow \text{AL}}]_+, \quad q_{w,4} = [s_w^{\text{ATR,AL} \rightarrow \text{LLu}} - e_w^{\text{AL}}]_+, \quad (22)$$

$$q_{w,5} = [s_w^{\text{VTRload}} - e_w^{\text{LLu}}]_+, \quad q_{w,6} = [s_w^{\text{ATR,LLl} \rightarrow \text{LP}} - e_w^{\text{LLl}}]_+, \quad (23)$$

where $[x]_+ = \max(0, x)$ and s and e denote stage start and end times. The route-total and route-maximum metrics are

$$Q_{\text{total}} = \sum_w \sum_{k=1}^6 q_{w,k}, \quad Q_{\text{max}} = \max_{w,k} q_{w,k}. \quad (24)$$

Processing cadence is reconstructed from the ordered start times of active full-flow and mixed-flow chamber batches. For adjacent-start intervals $d_1, \dots, d_m$, cadence variation is $\text{CV}(d) = \text{sd}(d) / \text{mean}(d)$.

The cadence-deviation sum is the sum of the absolute deviations used by the linear refinement objective after the discrete batch structure has been fixed. Table 2 reports the resulting instance-first means.

Table 2: Instance-first means in the matched, equal-total-budget stability-profile experiment.

Profile	Route-total	Route-maximum	Cadence CV	Cadence deviation
Full	2596.01	296.33	0.130	22.04
Quadratic objective (linear off)	2469.47	293.27	0.134	60.18
Stage2-off	5398.60	952.57	0.216	370.41

Relative to stage2-off, the full profile reduces route-total wait by 51.91%, route-maximum wait by 68.89%, cadence CV by 39.70%, and cadence deviation by 94.05%; all four comparisons remain significant after Holm adjustment. The linear objective also reduces cadence deviation by 63.38% relative to the quadratic profile. Its route-total, route-maximum, and cadence-CV differences from the quadratic profile are not Holm-adjusted significant. These estimates compare complete stability profiles under the same 600-s total budget, not a pure stage-two intervention applied to an identical stage-one incumbent: stage2-off can spend the full budget on the primary solve, whereas the two refinement profiles reserve up to 60 s. The results therefore identify the operational effect of the prespecified full two-stage configuration under a common total budget; they do not isolate a stage-two causal effect conditional on an equal primary search horizon.

5 Statistical protocol

The manufacturing instance is the unit of inference. Repeated solver and training realizations are first averaged within each instance and method or profile. Mean differences are reported with a nonparametric 95% bootstrap interval. Paired two-sided Wilcoxon signed-rank tests use the same instance pairs, and all planned contrasts within an analysis family are adjusted by the Holm procedure. Failure, incumbent, and optimality indicators are retained as binary outcomes. Primal-dual integral (PDI) is the primary anytime measure; lower values are better. Time-limited runs remain in PDI, time, incumbent, and failure summaries.

Specifically, if $G(t)$ is SCIP’s normalized primal-dual gap at time t ,

$$\text{PDI}(T) = \int_0^T G(t) dt. \quad (25)$$

Terminal gap is $G(T)$. Reported solution time is the sum of SCIP-reported primary and conditional-refinement solving times. It excludes model loading, solution serialization, and offline SPBS construction. The paired SPBS suite therefore estimates the downstream solve effect of loading a precomputed start; it is not an end-to-end comparison that charges start construction.

6 Solver, learning, and decoding configuration

All nominal methods receive the same precomputed, independently verified SPBS incumbent. Consequently, the 2700-run comparison isolates cut-selection and decoding policies under common primal guidance; it does not estimate an SPBS-policy interaction. The seven configurations are:

SCIP default Native SCIP cut selection with the shared SPBS incumbent.

ACS A four-score SCIP selector with direction, efficacy, integer-support, and objective-parallelism weights (0.25, 0, 0.50, 0.25).

HEM-13D The adapted 13-feature hierarchical policy with greedy decoding.

HEM-13D+Beam The identical trained HEM policy parameters with width-three beam decoding.

23D Features Only The reconstructed 23-feature policy with greedy decoding and no role-based reranking.

Structure+Greedy The trained proposed policy parameters and role-based completion with greedy decoding.

RDMCT-HBS The same trained proposed policy parameters and completion rule with width-three beam decoding.

Each candidate cut has 13 generic attributes: objective parallelism, efficacy, support, integer support, normalized violation, and the mean, maximum, minimum, and standard deviation of cut and objective coefficients. The ten added descriptors use only solver-visible variable types and optimization roles: coefficient-mass shares on SCIP binary, general-integer (including implied integer), nonzero-objective continuous, zero-objective auxiliary-continuous, and other variables, followed by bounded-variable mass, positive-coefficient mass, discrete-continuous coupling, dominant-family mass, and normalized family entropy. They are computed from SCIP type, objective, bound, column, and row-coefficient metadata; formulation-specific variable names are never parsed. Candidates are screened deterministically by descending efficacy, with original index as the tie breaker, to at most 128 rows; at most 16 rows are selected.

The 23-dimensional input passes through layer normalization, separate 13- and 10-dimensional projections, and a learned residual gate before a one-layer LSTM encoder and pointer-attention decoder. The recurrent hidden dimension is 128, with one attention glimpse and tanh exploration coefficient 5. A high-level policy chooses the selection fraction; a low-level pointer policy with an explicit end token generates an ordered, nonrepeating sequence. Beam width is three, ranked by cumulative joint log probability without length normalization. Structural completion retains at most $\lceil 0.25k \rceil$ policy anchors for target cardinality k and greedily fills from a pool of at most $2k$ candidates. Its quality, policy, role coverage, and representativeness weights are 0.35, 0.20, 0.25, and 0.20. The associated $(1 - 1/e)$ property applies only to the residual monotone-submodular completion problem with fixed anchors and pool, not to global cut selection.

Training uses five training LPs and five independent training seeds. For each seed, two workers jointly collect eight complete trajectories per epoch for 60 epochs with stochastic decoding; each trajectory has a 30-s/1024-MB SCIP budget, including at most 5 s of conditional refinement and one separation round at the root. A central synchronous Adam update uses an exponential-moving-average return baseline with coefficient 0.9. Low- and high-level policy learning rates are 10^{-4} and 5×10^{-4} ; the gradient norm is capped at 2, and both learning rates are multiplied by 0.96 every ten epochs. The return is negative SCIP PDI, with neither feature nor reward normalization. This is an A3C-inspired parallel hierarchical REINFORCE backbone with no learned critic and no asynchronous parameter update.

The terminal training epoch is prespecified rather than selected on benchmark performance. For each method family and seed, the retained epoch-60 parameter set must match the training split and seed and must have the required feature dimension and PDI reward. There is one admissible parameter set for each family-seed pair; no benchmark or test outcome enters parameter selection.

7 SPBS construction and acceptance

SPBS first fixes the canonical assignment, activation, and cleaning-epoch core in a temporary zero-objective feasibility MIP. The profiles then add or release load-lock-slot and resource-order decisions in the deterministic sequence shown in Table 3. Fractions are shares of the configured SPBS construction limit; every nonfinal attempt receives the smaller of its share and the remaining time, with a 5-s minimum when time remains, and the final profile receives the remainder. Each attempt stops at its first feasible solution.

Table 3: Deterministic SPBS release hierarchy for instances below 32 product wafers. The core comprises canonical assignment, activation, and cleaning epochs.

Profile	Fixed decisions	Limit share
Full	Core, load-lock slots, ATR/aligner, load-lock, and VTR orders	0.10
Relaxed-VTR	Core, slots, ATR/aligner and load-lock orders	0.15
Relaxed-LL	Core, slots and ATR/aligner orders	0.45
Fixed-slots	Core and load-lock slots	0.15
Structure-only	Core only	remainder

For at least 32 product wafers without a positive wait cap, the sequence is fixed-slots (0.50), structure-only (0.25), and an unrestricted final fallback. With a positive wait cap, it is structure-only (0.60) followed by the unrestricted fallback. The unrestricted profile removes even the structural fixings and uses only the remaining feasibility-search budget.

The selected physical schedule is reconstructed with secondary variables free in the full current model and is accepted only after SCIP’s complete original-space feasibility check. The production MILP receives only the accepted incumbent; no temporary fixing survives. Each cached start records the accepted profile and construction attempts and is reusable only when its full-model scope and exact LP SHA-256 match the current instance.

8 Hardware, software, and budgets

Experiments ran under Windows 11 Pro on an Intel Core Ultra 7 265KF (20 cores), 31.7 GB RAM, and an NVIDIA GeForce RTX 5060 Ti with 8 GB memory. The evaluation environment uses Python 3.12.12, SCIP 10.0.1 through PySCIPOpt 6.1.0, and PyTorch 2.11.0+cu130. The code does not impose a solver thread count, so no single-thread assumption is used in the analysis. Table 4 records the formal per-run limits.

All three stability profiles have the same 600-s total and 2048-MB memory budgets. Full and Quadratic-objective (linear-off) reserve up to 60 s for linear and quadratic refinement, respectively; Stage2-off assigns all 600 s to the primary solve. Their primary horizons are therefore unequal. Neither PDI/time nor the reconstructed route metrics identify a pure stage-two causal effect holding the stage-one incumbent and search horizon fixed. The matched comparison instead measures each complete profile under a common total budget, with route-wait and cadence outcomes reconstructed from saved event schedules under identical definitions.

Table 4: Formal per-run solver budgets.

Suite	Time (s)	Memory (MB)	Stage-two allocation
Validation	600	2048	up to 60 s
Nominal	600	2048	540 s primary + up to 60 s
Two-stage stability	600	2048	profile-specific; see text
Factorial DOE	600	2048	540 s primary + up to 60 s
SPBS	600	2048	540 s primary + up to 60 s
Sensitivity	600	2048	540 s primary + up to 60 s
Distribution shift	1200	4096	1140 s primary + up to 60 s

9 Independent SPBS warm-start experiment

Table 5 reports the controlled SPBS comparison. Automatic SPBS provides a valid incumbent in every run and reduces PDI by 7194.27 (26.19%). The matched instance-level effects on incumbent availability and PDI both remain significant after multiplicity adjustment. Both conditions start their 600-s timed production solve after any SPBS construction has finished; hence the comparison isolates the value of supplying the verified incumbent but does not include its offline generation cost.

Table 5: SPBS warm-start comparison over 48 instances and ten matched seeds.

Condition	Runs	Incumbent (%)	Optimal (%)	Mean PDI
Automatic SPBS	480	100.00	54.58	20278.78
No warm start	480	51.67	51.67	27473.05

The incumbent-rate difference is 48.33 percentage points (95% CI [34.58, 62.29]; Holm $p = 2.55 \times 10^{-5}$). The PDI difference is 7194.27 (95% CI [5145.78, 9317.61]; Holm $p = 3.46 \times 10^{-8}$). The optimal-rate and elapsed-time differences do not remain significant after Holm adjustment.

10 One-factor parameter sensitivity

Table 6 reports all eight prespecified settings. Each row contains ten default-SCIP runs and every run provides an independently valid incumbent. The experiment evaluates operational solvability under parameter perturbations; because it contains one method, it is not used for a cross- method robustness ranking.

11 Distribution-shift stress test

The OOD suite contains nine larger 40–64-wafer configurations. Its complete 729-run matrix has no duplicate key, execution error, solution-write error, or invalid saved solution. Table 7 reports the instance-first summaries. Every method attains 100% incumbent availability and 0% proven optimality under the common horizon.

Adaptive cut selection has the lowest OOD mean PDI. RDMCT-HBS has a mean terminal gap 0.78% below SCIP. None of the OOD pairwise contrasts remains significant after global Holm adjustment. The complete suite establishes the scalable feasibility of the shared SPBS–MILP–SCIP pipeline and complements the observed nominal-range mean anytime advantage reported in the main paper.

Table 6: One-factor sensitivity results for the 24-wafer 1:1 mixed-flow case.

Setting	Optimal (%)	Mean PDI	Mean time (s)
Nominal	20	40728.91	515.09
Motion scale 0.8	40	37126.93	494.55
Motion scale 1.2	0	42478.32	539.79
Processing scale 0.8	60	33961.82	459.63
Processing scale 1.2	10	38663.67	508.40
Cleaning interval 3	10	38429.95	517.26
Cleaning interval 10	40	28899.41	497.91
Vacuum-zone dummy-wafer inventory 10	20	38152.52	484.80

Table 7: Complete OOD comparison over nine larger configurations.

Method	Runs	Mean PDI	Mean gap (%)
SCIP	27	73670.99	171.35
Adaptive cut selection	27	70915.54	160.78
HEM-13D	135	73003.59	165.88
HEM-13D+Beam	135	73165.58	167.04
23D Features Only	135	74187.22	169.96
Structure+Greedy	135	73507.76	168.54
RDMCT-HBS	135	73919.15	170.02

12 Factorial-model diagnostics

The factorial design contains 60 complete cells and five solver seeds per cell. The prespecified PDI model has $R^2 = 0.9980$. Wafer count and recipe composition exhibit strong interactions, whereas all prespecified processing- scale main-effect and interaction confidence intervals include zero. Figure 1 provides the residual and influence diagnostics.

13 Reproducibility record

The reproducibility package preserves the instance manifest, solver and training seeds, method configuration, time and memory limits, trained-model metadata, raw row records, saved incumbent solutions, independently derived manufacturing metrics, and machine-readable statistical summaries. Every formal suite is accepted only when its expected matrix is complete, duplicate keys and execution errors are absent, saved solutions are covered one-to-one by independent validation, and the statistical input count equals the frozen raw count.

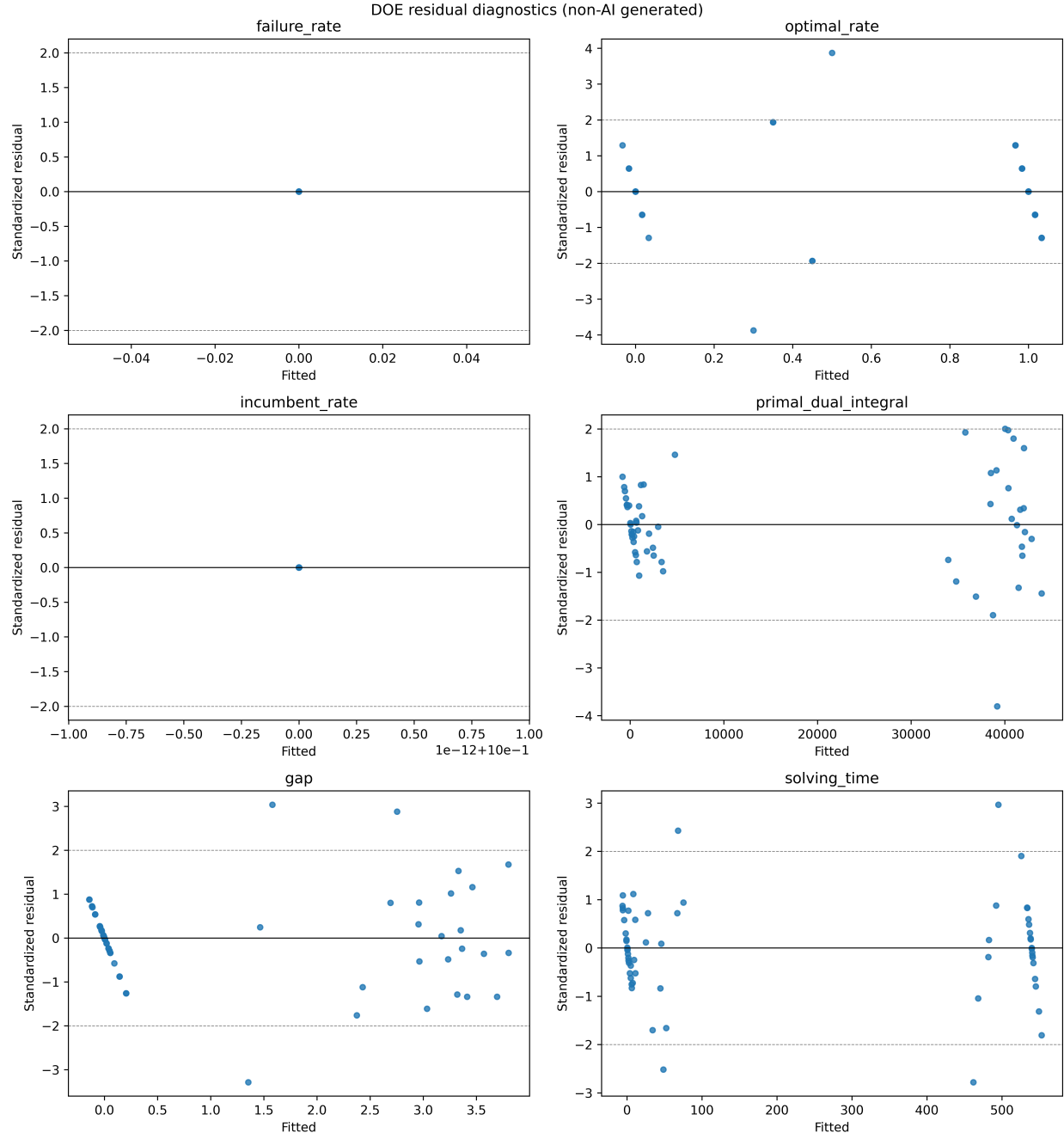


Figure 1: Diagnostic plots for the prespecified factorial PDI model fitted to the complete 300-run design.